

Foundation Mathematics 1017SCG
Week 6 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a) The start of April is **3 months** after the start of the year ($t = 3$).

$$\begin{aligned}N &= N_0 e^{kt} \\100 &= 80 e^{k \times 3} \\\frac{100}{80} &= \frac{80 e^{3k}}{80} \\\frac{5}{4} &= e^{3k} \\\ln\left(\frac{5}{4}\right) &= \ln(e^{3k}) \\\ln\left(\frac{5}{4}\right) &= 3k \ln(e) \\\ln\left(\frac{5}{4}\right) &= 3k \\\frac{\ln\left(\frac{5}{4}\right)}{3} &= \frac{3k}{3} \\k &= 0.0744\end{aligned}$$

- (b) The start of August is **7 months** after the start of the year ($t = 7$).

$$\begin{aligned}N &= N_0 e^{kt} \\&= 80 e^{0.0744 \times 7} \\&= 134.67\end{aligned}$$

There will be approximately 135 fish in the pond at the start of August.

- (c)

$$\begin{aligned}N &= N_0 e^{kt} \\220 &= 80 e^{0.0744 \times t} \\\frac{220}{80} &= \frac{80 e^{0.0744 \times t}}{80} \\\frac{11}{4} &= e^{0.0744t} \\\ln\left(\frac{11}{4}\right) &= \ln(e^{0.0744t}) \\\ln\left(\frac{11}{4}\right) &= 0.0744t \ln(e) \\\ln\left(\frac{11}{4}\right) &= 0.0744t \\\frac{\ln\left(\frac{11}{4}\right)}{0.0744} &= \frac{0.0744t}{0.0744} \\t &= 13.60\end{aligned}$$

The number of fish in the pond is predicted to reach 220 by mid-February the following year.

2. (a)

$$\begin{aligned}N &= N_0 e^{kt} \\10 &= 15e^{k \times 60} \\ \frac{10}{15} &= \frac{15e^{60k}}{15} \\ \frac{2}{3} &= e^{60k} \\ \ln\left(\frac{2}{3}\right) &= \ln(e^{60k}) \\ \ln\left(\frac{2}{3}\right) &= 60k \ln(e) \\ \ln\left(\frac{2}{3}\right) &= 60k \\ \frac{\ln\left(\frac{2}{3}\right)}{60} &= \frac{60k}{60} \\ k &= -0.0068\end{aligned}$$

- (b) To determine the half-life (the time required for half the radioactive substance to decay), we will determine how long it takes for the initial 15 grams to decay to 7.5 grams.

$$\begin{aligned}N &= N_0 e^{kt} \\7.5 &= 15e^{-0.0068 \times t} \\ \frac{7.5}{15} &= \frac{15e^{-0.0068 \times t}}{15} \\ \frac{7.5}{15} &= e^{-0.0068t} \\ \ln\left(\frac{1}{2}\right) &= \ln(e^{-0.0068t}) \\ \ln\left(\frac{1}{2}\right) &= -0.0068t \ln(e) \\ \ln\left(\frac{1}{2}\right) &= -0.0068t \\ \frac{\ln\left(\frac{1}{2}\right)}{-0.0068} &= \frac{-0.0068t}{-0.0068} \\ t &= 101.93\end{aligned}$$

The half-life of the radioactive substance is approximately 102 years.

3. (a)

$$\begin{aligned}N &= N_0 e^{kt} \\25.7 &= 19e^{k \times 21} \\\frac{25.7}{19} &= \frac{19e^{21k}}{19} \\\frac{25.7}{19} &= e^{21k} \\\ln\left(\frac{25.7}{19}\right) &= \ln(e^{21k}) \\\ln\left(\frac{25.7}{19}\right) &= 21k \ln(e) \\\ln\left(\frac{25.7}{19}\right) &= 21k \\\frac{\ln\left(\frac{25.7}{19}\right)}{21} &= \frac{21k}{21} \\k &= 0.014383\end{aligned}$$

In the solution, rounding has only occurred in the final line of working. If you rounded multiple times throughout the question, you may find that your value of k is slightly different.

(b)

$$\begin{aligned}N &= 19e^{0.014383 \times 30} \\&= 29.25\end{aligned}$$

Australia's population is predicted to be 29.25 million at the start of 2030.

(c) To make this question a little easier, the value of k has not been substituted into the equation until the final step.

$$\begin{aligned}N &= N_0 e^{kt} \\40 &= 19e^{kt} \\\frac{40}{19} &= \frac{19e^{kt}}{19} \\\frac{40}{19} &= e^{kt} \\\ln\left(\frac{40}{19}\right) &= \ln(e^{kt}) \\\ln\left(\frac{40}{19}\right) &= kt \ln(e) \\\ln\left(\frac{40}{19}\right) &= kt \\\frac{\ln\left(\frac{40}{19}\right)}{k} &= \frac{kt}{k} \\t &= \frac{\ln\left(\frac{40}{19}\right)}{0.014383} \\t &= 51.76\end{aligned}$$

It is predicted that Australia's population will reach 40 million in the year 2051.

4. (a)

$$\begin{aligned}P &= 400 \times 10^{0.08 \times 3} \\&= 695.12\end{aligned}$$

After three months there will be approximately 695 fish in the dam.

(b)

$$\begin{aligned}P &= 400 \times 10^{0.08 \times 8} \\&= 1746.06\end{aligned}$$

After eight months there will be approximately 1746 fish in the dam.

(c)

$$\begin{aligned}P &= 400 \times 10^{0.08t} \\5000 &= 400 \times 10^{0.08t} \\\frac{5000}{400} &= \frac{400 \times 10^{0.08t}}{400} \\\frac{25}{2} &= 10^{0.08t} \\\log_{10}\left(\frac{25}{2}\right) &= \log_{10}(10^{0.08t}) \\\log_{10}\left(\frac{25}{2}\right) &= 0.08t \times \log_{10}(10) \\\log_{10}\left(\frac{25}{2}\right) &= 0.08t \\\frac{\log_{10}\left(\frac{25}{2}\right)}{0.08} &= \frac{0.08t}{0.08} \\t &= 13.71\end{aligned}$$

It will take a little under 14 months for the fish population in the dam to reach 5000.

(d) As the fish population can be modelled using an exponential **growth** model, the number of fish will continue to increase. Therefore, the minimum number of fish in the dam is 400 which occurs when the fish were first introduced into the dam ($t = 0$).

Question 5-9 on the tutorial sheet ask you to solve trigonometric equations which require using the special triangles and unit circle. As it is difficult to include these diagrams in the solutions, it is best to ask questions and receive help for these questions during the weekly tutorial.

5. (a)

$$\sin(\theta) = \frac{1}{2}$$

The first solution can be obtained from the special triangles: $\theta = 30^\circ$. Another solution can be found in the second quadrant (as sin is positive in the second quadrant). This gives $\theta = 150^\circ$. Therefore the two solutions are $\theta = 30^\circ$ and $\theta = 150^\circ$.

(b)

$$\tan(\theta) = \frac{1}{\sqrt{3}}$$

The first solution can be obtained from the special triangles: $\theta = 30^\circ$. Another solution can be found in the third quadrant (as tan is positive in the third quadrant). This gives $\theta = 210^\circ$. Therefore the two solutions are $\theta = 30^\circ$ and $\theta = 210^\circ$.

(c)

$$\cos(\theta) = \frac{1}{\sqrt{2}}$$

The first solution can be obtained from the special triangles: $\theta = 45^\circ$. Another solution can be found in the fourth quadrant. This gives $\theta = 315^\circ$. Therefore the two solutions are $\theta = 45^\circ$ and $\theta = 315^\circ$.

(d)

$$\sin(\theta) = \frac{\sqrt{3}}{2}$$

The first solution can be obtained from the special triangles: $\theta = 60^\circ$. Another solution can be found in the second quadrant. This gives $\theta = 120^\circ$. Therefore the two solutions are $\theta = 60^\circ$ and $\theta = 120^\circ$.

6. (a)

$$\tan(\theta) = \sqrt{3}$$

The first solution can be obtained from the special triangles: $\theta = \frac{\pi}{3}$. Another solution can be found in the third quadrant. This gives $\theta = \frac{4\pi}{3}$. Therefore the two solutions are $\theta = \frac{\pi}{3}$ and $\theta = \frac{4\pi}{3}$.

(b)

$$\begin{aligned} 10 \cos(\theta) &= 5 \\ \frac{10 \cos(\theta)}{10} &= \frac{5}{10} \\ \cos(\theta) &= \frac{1}{2} \end{aligned}$$

The first solution can be obtained from the special triangles: $\theta = \frac{\pi}{3}$. Another solution can be found in the fourth quadrant. This gives $\theta = \frac{5\pi}{3}$. Therefore the two solutions are $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$.

(c)

$$\begin{aligned} \sqrt{2} \sin(\theta) &= 1 \\ \frac{\sqrt{2} \sin(\theta)}{\sqrt{2}} &= \frac{1}{\sqrt{2}} \\ \sin(\theta) &= \frac{1}{\sqrt{2}} \end{aligned}$$

The first solution can be obtained from the special triangles: $\theta = \frac{\pi}{4}$. Another solution can be found in the second quadrant. This gives $\theta = \frac{3\pi}{4}$. Therefore the two solutions are $\theta = \frac{\pi}{4}$ and $\theta = \frac{3\pi}{4}$.

(d)

$$\begin{aligned} \sin(\theta) &= \cos(\theta) \\ \frac{\sin(\theta)}{\cos(\theta)} &= \frac{\cos(\theta)}{\cos(\theta)} \\ \tan(\theta) &= 1 \end{aligned}$$

The first solution can be obtained from the special triangles: $\theta = \frac{\pi}{4}$. Another solution can be found in the third quadrant. This gives $\theta = \frac{5\pi}{4}$. Therefore the two solutions are $\theta = \frac{\pi}{4}$ and $\theta = \frac{5\pi}{4}$.

7. (a)

$$\sin(\theta) = -\frac{1}{2}$$

We will first start by solving $\sin(y) = \frac{1}{2}$ (the positive case). This gives $y = 30^\circ$.

However, as we are solving for the negative case, we will look for solutions in the third and fourth quadrants. Therefore the two solutions are $\theta = 210^\circ$ and $\theta = 330^\circ$.

(b)

$$\cos(\theta) = -\frac{\sqrt{3}}{2}$$

We will first start by solving $\cos(y) = \frac{\sqrt{3}}{2}$ (the positive case). This gives $y = 30^\circ$.

However, as we are solving for the negative case, we will look for solutions in the second and third quadrants. Therefore the two solutions are $\theta = 150^\circ$ and $\theta = 210^\circ$.

8. (a)

$$\tan(\theta) = -\frac{1}{\sqrt{3}}$$

We will first start by solving $\tan(y) = \frac{1}{\sqrt{3}}$ (the positive case). This gives $y = \frac{\pi}{6}$.

However, as we are solving for the negative case, we will look for solutions in the second and fourth quadrants. Therefore the two solutions are $\theta = -\frac{\pi}{6}$ (fourth quadrant) and $\theta = \frac{5\pi}{6}$ (second quadrant).

It is important to note that while $\theta = \frac{11\pi}{6}$ (fourth quadrant) is a solution to $\tan(\theta) = -\frac{1}{\sqrt{3}}$, we were asked to find solutions such as $-\pi \leq \theta \leq \pi$.

(b)

$$\cos(\theta) = -\frac{1}{\sqrt{2}}$$

We will first start by solving $\cos(y) = \frac{1}{\sqrt{2}}$ (the positive case). This gives $y = \frac{\pi}{4}$.

However, as we are solving for the negative case, we will look for solutions in the second and third quadrants. Therefore the two solutions are $x = -\frac{3\pi}{4}$ and $x = \frac{3\pi}{4}$.

9. (a)

$$5 \tan(x) - 2 = 0$$

$$5 \tan(x) = 2$$

$$\tan(x) = \frac{2}{5}$$

In this question, a solution cannot be obtained from the special triangles.

$$\tan(x) = \frac{2}{5}$$

$$x = \tan^{-1}\left(\frac{2}{5}\right)$$

$$x = 0.38$$

As $\tan$ is positive, we can look for solutions in the first and third quadrant. However, solutions in the first quadrant will not satisfy $\pi \leq x \leq 2\pi$. Therefore the only solution is $x = 3.52$ (third quadrant).

(b)

$$10 \cos(x) + 9 = 0$$

$$10 \cos(x) = -9$$

$$\cos(x) = -\frac{9}{10}$$

We will first start by solving $\cos(y) = \frac{9}{10}$ (the positive case). In this question, a solution cannot be obtained from the special triangles.

$$\cos(y) = \frac{9}{10}$$

$$y = \cos^{-1}\left(\frac{9}{10}\right)$$

$$y = 0.45$$

As we are trying to solve the negative case (where $\cos$ is negative), we can look for solutions in the second and third quadrant. However, solutions in the second quadrant will not satisfy $\pi \leq \theta \leq 2\pi$. Therefore the only solution is $\theta = 3.59$ (third quadrant).

10.

(a)

$$f(x) = 2x - 5$$

$$\begin{aligned} f(6) &= 2 \times 6 - 5 \\ &= 12 - 5 \\ &= 7 \end{aligned}$$

(b)

$$\begin{aligned} f(-3) &= 2 \times (-3) - 5 \\ &= -6 - 5 \\ &= -11 \end{aligned}$$

11.

(a)

$$g(x) = -x^2 + 3x$$

$$\begin{aligned} g(4) &= -(4)^2 + 3 \times (4) \\ &= -16 + 12 \\ &= -4 \end{aligned}$$

(b)

$$\begin{aligned} g(-2) &= -(-2)^2 + 3 \times (-2) \\ &= -4 - 6 \\ &= -10 \end{aligned}$$

12.

$$f(x) = 3x^2 + 2x - 1$$

(a)

$$\begin{aligned}f(4) &= 3 \times 4^2 + 2 \times 4 - 1 \\&= 3 \times 16 + 2 \times 4 - 1 \\&= 48 + 8 - 1 \\&= 55\end{aligned}$$

(b)

$$\begin{aligned}f(-5) &= 3 \times (-5)^2 + 2 \times (-5) - 1 \\&= 3 \times 25 + 2 \times -5 - 1 \\&= 75 - 10 - 1 \\&= 64\end{aligned}$$

(c)

$$\begin{aligned}f(0) &= 3 \times 0^2 + 2 \times 0 - 1 \\&= 3 \times 0 + 2 \times 0 - 1 \\&= 0 + 0 - 1 \\&= -1\end{aligned}$$

(d)

$$\begin{aligned}f(3a) &= 3 \times (3a)^2 + 2 \times 3a - 1 \\&= 3 \times 9a^2 + 2 \times 3a - 1 \\&= 27a^2 + 6a - 1\end{aligned}$$

(e)

$$\begin{aligned}f(b+1) &= 3 \times (b+1)^2 + 2 \times (b+1) - 1 \\&= 3 \times (b^2 + 2b + 1) + 2 \times (b+1) - 1 \\&= 3b^2 + 6b + 3 + 2b + 2 - 1 \\&= 3b^2 + 8b + 4\end{aligned}$$

13.

$$f(x) = \sqrt{x}$$

- (a) For the function $f(x)$, $x = -3$ is NOT included in the domain as we cannot evaluate the square root of a negative number (unless we use complex numbers).
- (b) For the function $f(x)$, $x = 4$ is included in the domain as when it is substituted into the function, it returns a sensible answer. $f(4) = \sqrt{4} = 2$.
- (c) The domain of the function $f(x)$ is $x \geq 0$.

14.

$$f(x) = \frac{1}{x-2}$$

- (a) For the function $f(x)$, $x = 2$ is NOT included in the domain as we cannot divide by zero.

- (b) For the function $f(x)$, $x = 5$ is included in the domain as when it is substituted into the function, it returns a sensible answer. $f(5) = \frac{1}{5-2} = \frac{1}{3}$.
- (c) The domain of the function $f(x)$ is all values of x except $x = 2$.

15.

$$f(x) = \log_{10}(x)$$

- (a) For the function $f(x)$, $x = 0$ is NOT included in the domain as we cannot evaluate the log of zero.
- (b) For the function $f(x)$, $x = -10$ is NOT part of the domain as we cannot evaluate the log of a negative number.
- (c) For the function $f(x)$, $x = 1$ is included in the domain as when it is substituted into the function, it returns a sensible answer. $f(1) = \log_{10}(1) = 0$.
- (d) The domain of the function $f(x)$ is $x > 0$.

16.

$$f(x) = x^2$$

- (a) For the function $f(x)$, -9 is NOT included in the range as when we square a number, it will always return a positive number (or zero).
- (b) For the function $f(x)$, 16 is included in the range as $f(4) = 4^2 = 16$. Therefore it is possible to have 16 as the output of the function and therefore is included in the range of the function.
- (c) The domain of the function $f(x)$ is all values of x . The range of the function $f(x)$ is $f(x) \geq 0$.